

Working with VeriFact-BHC dataset (100 MIMIC-III patients).

Goal: Generate Brief Hospital Course summaries using baseline (zero-shot) and RAG pipeline, evaluate against human-written BHC.

Completed

✓ Project Planning & Setup

- Defined research question, baseline (zero-shot), and proposed method (RAG + structured template)
- Submitted project proposal (Week 4)
- Completed literature review — identified key papers (Van Veen et al., Alkhalaf et al., VeriFact/Chung et al., PDSQI-9/Croxford et al., CLI-RAG, CLEAR/Lopez et al.)
- Created project guide v4 (definitive reference document)
- Set up group GitHub repo (`bmi702_group_1`) with lean folder structure
- Set up local conda environment (`702-project`) on Mac with PyTorch, Transformers, FAISS, sentence-transformers, rouge-score, bert-score
- Created `requirements_colab.txt` for GPU inference on Colab

Step 1 — Data Exploration

Notebook: `notebooks/01_data_exploration.ipynb`

Dataset: MIMIC-III-Ext-VeriFact-BHC v1.0.0 (PhysioNet)

Data access: Symlinked from central datasets folder → `data/verifact-bhc/`

Key findings:

Metric	Value
Patients	100
Hospital admissions	125 (25 patients have prior admissions)
Total clinical notes	4,787
Note categories	14 (Nursing, Physician, Radiology, Respiratory, ECG, etc.)
Notes per patient	Median 25, range 10–314
Tokens per patient	Median 17,562, range 5,720–232,852
Patients exceeding 32K tokens	31/100
Patients exceeding 128K tokens	9/100
Human-written BHC length	Median 513 tokens, range 107–1,905
Total propositions	13,070
Human BHC propositions	8,625 (5,648 claims + 2,977 sentences)

Proposition ground truth distribution:

Label	Overall	Human BHC	LLM BHC
Supported	73.6%	62.6%	95.0%
Not Addressed	17.6%	25.6%	1.9%
Not Supported	8.8%	11.8%	3.1%

Key insight: Human-written BHCs have 25.6% "Not Addressed" because clinicians had access to information beyond MIMIC-III (bedside conversations, notes from other services). Our pipeline will only have access to MIMIC-III notes, so we should expect factual support rates closer to the LLM column (~95% Supported). ROUGE/BERTScore against the human BHC will be moderate because the human wrote about things our model can't know — this is an expected limitation, not a failure.

Pipeline design implications:

- 31/100 patients exceed 32K tokens → baseline will hit context window limits for many patients
- 14 note categories → RAG must handle diverse clinical documentation types
- Median 25 notes per patient → genuine multi-document fragmentation (this is what makes RAG valuable)

- Top note categories by volume: Nursing (1,228), Physician (1,223), Radiology (1,134)
- Longest note types: Discharge summaries (~3,063 tokens), Physician notes (~1,778 tokens)

Processed data saved to: `data/verifact-bhc/processed/`

- `notes.parquet` — 4,787 notes with token length columns
- `human_bhcs.parquet` — 100 gold standard BHC targets
- `propositions_with_gt.parquet` — 13,070 propositions with human ground truth labels

Step 2 — Baseline Pipeline

Notebook: `notebooks/02_baseline.ipynb`

Environment: Google Colab Pro (A100 GPU)

Model: Mistral 7B Instruct v0.3, 4-bit quantized (~4.1GB VRAM)

Setup:

- Zero-shot prompting: concatenate all notes chronologically → simple instruction → generate BHC
- No retrieval, no structured template, no examples
- 32K context window; patients exceeding this were truncated (keeps earliest notes)

Baseline Results:

Metric	Value
ROUGE-1 F1	0.339 (±0.075)
ROUGE-2 F1	0.084 (±0.040)
ROUGE-L F1	0.165 (±0.042)
BERTScore F1	0.819 (±0.014)
Patients truncated	42/100
Generated BHC length	Mean 441 tokens
Human BHC length	Mean 743 tokens
Runtime	65.6 minutes (100 patients)

Truncation impact on quality:

Metric	Truncated (n=42)	Not truncated (n=58)
ROUGE-1 F1	0.314 (± 0.074)	0.358 (± 0.070)
ROUGE-L F1	0.148 (± 0.032)	0.177 (± 0.044)
BERTScore F1	0.814 (± 0.013)	0.823 (± 0.014)
Avg input tokens	105,563	17,719

Results saved to Google Drive:

- `baseline_results.parquet` — generated BHCs for all 100 patients
- `baseline_eval_results.parquet` — evaluation scores per patient

Step 3 — RAG Pipeline

Notebook: `notebooks/03_rag_pipeline.ipynb`

Environment: Google Colab Pro (A100 GPU)

LLM: Mistral 7B Instruct v0.3, 4-bit quantized (same as baseline)

Embedding model: BGE-M3 (1024-dim, same model used in VeriFact paper)

RAG pipeline:

1. Chunked 4,787 clinical notes → 23,663 chunks (~300 tokens each, 50-token overlap)
2. Embedded all chunks with BGE-M3 (221s on A100)
3. Built per-patient FAISS indices (100 isolated indices, cosine similarity)
4. Three retrieval strategies tested iteratively:
 - **RAG v1 (single query, ~11K budget):** One generic "hospital course" query, top-40 chunks
 - **RAG v2 (multi-query, ~10K budget):** Six clinically-motivated queries (admission, diagnoses, workup, treatment, course, discharge), deduplicated
 - **RAG v3 (multi-query, ~28K budget):** Same six queries with token budget matched to baseline's context window

Iterative design process:

Version	Change	Rationale
v1 → v2	Single query → six clinical queries	Single query retrieved templated content (med lists, ICU order sets) instead of clinical narrative. Multi-query mirrors how physicians organize hospital courses — ensuring diversity across all BHC components.
v2 → v3	10K token budget → 28K token budget	Discovered RAG was using 1/3 the context baseline had (~10K vs ~31K tokens). Matching the budget gave RAG a fair comparison. Large patients (e.g., 356K tokens) went from 11K to 28K retrieved tokens.

Step 4 — Evaluation: Baseline vs RAG

Automated metrics (full cohort, n=100):

Metric	Baseline	RAG v1 (single-Q, 11K)	RAG v2 (multi-Q, 10K)	RAG v3 (multi-Q, 28K)
ROUGE-1 F1	0.339	0.339	0.335	0.334
ROUGE-2 F1	0.084	0.079	0.081	0.084
ROUGE-L F1	0.165	0.163	0.162	0.168
BERTScore F1	0.819	0.818	0.818	0.820
Avg context	~31K (trunc)	~11K	~10K	~18K

RAG v3 is the first variant to match or exceed baseline on automated metrics — using 42% less context.

Subgroup analysis — Baseline vs RAG v3:

Metric	Baseline (trunc, n=42)	RAG v3 (trunc, n=42)	Baseline (no trunc, n=58)	RAG v3 (no trunc, n=58)
ROUGE-1 F1	0.314	0.315	0.358	0.350
ROUGE-L F1	0.148	0.147	0.177	0.173
BERTScore F1	0.814	0.813	0.823	0.822

Proposition-level fact recall (threshold=0.60):

Using clinician-verified "Supported" propositions from the VeriFact dataset (5,396 verified facts across 100 patients) as a factual checklist. For each proposition, semantic similarity (BGE-M3 cosine) determines whether the generated BHC captures the fact.

Metric	Baseline	RAG v2	RAG v3
Mean fact recall	73.5%	72.2%	72.5%
Total facts captured	3,735/5,396	3,662/5,396	3,711/5,396

Subgroup	Baseline recall	RAG v3 recall	Δ
Truncated (n=42)	70.0%	69.8%	-0.3%
Not truncated (n=58)	76.0%	74.5%	-1.5%

Per-patient fact recall (RAG v3 vs baseline): Truncated — 19/42 improved, 20/42 worsened, 3/42 tied. Not truncated — 19/58 improved, 32/58 worsened, 7/58 tied.

Qualitative findings:

- **Patient 72940 (CABG, 37K tokens):** Baseline hallucinated wrong admission diagnosis ("aortoiliac disease"); RAG correctly identified procedure with specific graft details (LIMA-LAD, SVG-OM) matching the human BHC. ROUGE-L: baseline 0.113 → RAG 0.240.
- **Patient 61157 (SAH, 86K tokens):** Baseline captured narrative arc well from early notes; RAG produced longer but vaguer summary. ROUGE-L: baseline 0.201 → RAG 0.153.

Key Findings

Finding 1 — RAG matches baseline quality with significantly less context.

RAG v3 achieves equivalent or slightly better automated metric scores (ROUGE-L 0.168 vs 0.165, BERTScore 0.820 vs 0.819) while using 42% less context (~18K vs ~31K tokens). RAG selects relevant content effectively.

Finding 2 — Retrieval strategy matters.

Single-query RAG retrieved templated content; multi-query with clinically-motivated queries improved diversity. Token budget matching was critical — initial RAG used 1/3 baseline's context, creating an unfair comparison.

Finding 3 — Chronological truncation is a surprisingly strong baseline.

Keeping the earliest notes preserves admission, initial workup, and early assessments — the backbone of any BHC. Later notes (often repetitive daily progress notes) contribute less to BHC quality than expected.

Finding 4 — Standard metrics are insufficient for clinical summarization evaluation.

ROUGE and BERTScore cannot distinguish factual accuracy from plausible-sounding text. Proposition-level recall showed similar patterns across approaches (~73%). Qualitative review revealed hallucination differences invisible to automated metrics.

Finding 5 — Aligns with the literature.

Alkhalaf et al. (2024) showed RAG improving accuracy measured by verification, not ROUGE. VeriFact (Chung et al. 2025) demonstrated LLM text achieves 95% factual support vs 62.6% for human BHCs. The disconnect between automated metrics and clinical quality is a known, important problem.
